Fiscal Year:	FY 2008	**Task Last Updated:**	FY 06/02/2008
PI Name:	Neitzel, G. Paul Ph.D.		
Project Title:	Novel Applications of Permanent Nonwetting		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Interfacial phenomena		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03-C
Start Date:	01/01/2005	**End Date:**	12/31/2008
No. of Post Docs:		**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: received notification 8/2006 that funding reinstated through 12/31/2008, per J. Cohen (HQ)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNC05GA33G		
Performance Goal No.:			
Performance Goal Text:			
	Long-duration space-exploration missions require that new solutions be sought for the resolution of issues for which short-mission solutions are unsatisfactory. Up-mass constraints require that these new solutions be mass-efficient and robust. A class of new methods that falls into this category are embodied in so-called "lab-on-a-chip" devices for the analysis of minute volumes of liquid. Many of these devices rely on the use of hard-wired channels or "capillaries," through which liquid is forced. Since the pressure difference required to drive liquid at a given flow rate through a pipe varies as R^-4, this becomes increasingly difficult as capillary radius R decreases. Other recent work has used surface forces such as thermocapillarity to move liquid along a surface, although in contact with it, the transport being slowed by viscous action through the no-slip condition. We have shown that permanent nonwetting of a solid by a normally wetting liquid may be achieved by providing relative tangential motion between the liquid and solid surfaces in		

Task Description:	question. This may be accomplished by physically moving one surface relative to the other (e.g., sliding a solid surface past a liquid droplet) or by employing a mechanism such as thermocapillarity to cause the liquid surface to move tangentially. In either case, a lubricating film of surrounding gas is drawn into the space between the two surfaces, causing them to remain far enough apart that attractive van der Waals forces are ineffective. The proposed research intends to demonstrate that liquid droplets may be levitated above (or between) solid surfaces and moved from one position to another using directional heating to propel a droplet in a nearly frictionless manner, thus avoiding the capillary-pumping restrictions. The heating is done optically using infrared radiation that is controllable to a degree necessary to effect precise transport of the droplets. To eliminate undesirable sample evaporation, and to permit the transport of aqueous solutions that are resistant to the use of thermocapillarity, we shall employ liquid encapsulation of aqueous droplets in less-volatile silicone oils, which have been shown to exhibit excellent thermocapillary transport properties. Three side benefits of encapsulation of this type are: i) minimization of sample-to-sample contamination (crucial to such lab-on-a-chip applications as DNA testing); ii) minimization of evaporation losses of volatile (and often valuable) aqueous samples; and iii) the provision of a less-viscous core within a silicone-oil droplet enabling more vigorous surface motion, and hence, enhanced transport, using smaller surface-temperature gradients. The proposed scheme has positive implications for the design of a new class of lab-on-a-chip architectures for use in space-exploration applications. In fact, the method will perform better in a microgravity environment than in a terrestrial one, due to the absence of body forces that may require the application of additional pumping force to overcome.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research identifies a novel droplet transport scheme with possible applications to lab-on-a-chip devices. This new scheme permits more rapid transport of droplets due to extremely small frictional forces and virtually no droplet-to-droplet contamination due to levitation.
Task Progress:	Failure Modes for Nonwetting Systems--We have investigated failure modes associated with a nonwetting heated droplet squeezed against a cooled glass plate for a number of conditions. Failure occurs through one of two modes: i) film failure, in which the lubricating gas film undergoes a shape instability, allowing the liquid and solid surfaces to come into contact; or ii) pinning failure, in which the contact line pinning the liquid droplet to its support fails. Friction of a Sliding Droplet--The coefficient of static friction for a thermocapillary-nonwetting droplet pressed against a plate is zero, owing to the symmetry of the lubricating film driven by thermocapillarity. To quantify such frictional forces, we have measured the force on a droplet held fixed above a non-wetted, rotating disk against which the droplet is being pressed. Optical Droplet Levitation and Translation--We have optically (using IR radiation from a CO2 laser) levitated a droplet above a solid surface, translating the droplet by moving the location of the incident radiation. Modeling Efforts--We have made significant progress in numerically modeling liquid and gas flows for both thermocapillary and isothermal nonwetting droplets in two dimensions. In two-dimensions, an interesting blockage effect exists for the isothermal sliding droplet that is not present in an actual three-dimensional experiment. We are presently engaged in three-dimensional simulations using the commercial code COMSOL. Nanoliter Compound-Droplet Formation--We have been successful in creating compound droplets with an inner aqueous droplet of O(10 nl) surrounded by silicone oil with a total compound-droplet volume of O(100 nl). We have modified the device to be able to more effective vary the water/oil volume ration and tests are presently underway.
Bibliography Type:	Description: (Last Updated: 12/31/2020)
Abstracts for Journals and Proceedings	Neitzel GP. "Non-coalescence, Non-Wetting and Applications." Invited presentation at the Pan American Advanced Studies Institute on Interfacial Fluid Dynamics: From Theory to Application, Mar del Plata, Argentina, August 6 – 17, 2007. Pan American Advanced Studies Institute on Interfacial Fluid Dynamics: From Theory to Application, Mar del Plata, Argentina, August 6 – 17, 2007. , Aug-2007
Abstracts for Journals and Proceedings	Smith MK, Neitzel GP. "Multiscale modeling in the numerical computation of isothermal nonwetting." Presented at Session ND Rising and Spreading of Drops and Bubbles, the 58th Meeting of the Division of Fluid Dynamics of the American Physical Society, Chicago, IL, November 20-22, 2005. 58th Meeting of the Division of Fluid Dynamics of the American Physical Society, Chicago, IL, November 20-22, 2005. Program and abstracts. , Nov-2005
Abstracts for Journals and Proceedings	Carnasciali MI, Neitzel GP. "Frictional forces associated with isothermal non-wetting." Presented at Session AC Microfluidics: Drops, Bubbles, and Wetting, the 58th Meeting of the Division of Fluid Dynamics of the American Physical Society, Chicago, IL, November 20-22, 2005. 58th Meeting of the Division of Fluid Dynamics of the American Physical Society, Chicago, IL, November 20-22, 2005. Program and abstracts. , Nov-2005
Abstracts for Journals and Proceedings	Neitzel GP, Nagy PT. "Failure loading on a thermocapillary nonwetting droplet." Presented at Session AC Microfluidics: Drops, Bubbles, and Wetting, the 58th Meeting of the Division of Fluid Dynamics of the American Physical Society, Chicago, IL, November 20-22, 2005. 58th Meeting of the Division of Fluid Dynamics of the American Physical Society, Chicago, IL, November 20-22, 2005. Program and abstracts. , Nov-2005
Articles in Peer-reviewed Journals	Kuo C-W, Chen J-C, Neitzel GP. "Numerical simulation of isothermal nonwetting." International Journal for Numerical Methods in Fluids 2007 Jan 20;53(2):257-75. http://dx.doi.org/10.1002/fld.1280 , Jan-2007
Dissertations and Theses	Carnasciali M-I. "Kinetic friction of nonwetting drops." Ph.D. Thesis, Georgia Institute of Technology, Atlanta, GA, April 2008. , Apr-2008

Dissertations and Theses	Nagy PT. "Investigation of nonwetting system failure and system integration." Ph.D. thesis, mechanical engineering, Georgia Institute of Technology, December, 2006. , Dec-2006